

Land Use Land Cover Classification

Using Landsat 8/9 Imagery and Deep Learning

Submitted by:

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How to run the code

The notebook is designed to be run on a cloud GPU like Kaggle (T4/P100) or Google Colab.

1. API Keys: Have your USGS M2M API credentials ready. The notebook will ask for them to download the satellite imagery.
2. Setup: If you are running this somewhere other than Kaggle, you will need to install the dependencies:
3. Bash
4. `pip install torch torchvision segmentation-models-pytorch rasterio albumentations scikit-learn geopandas matplotlib tqdm`
5. Execution: Just run the notebook from top to bottom. It will handle the downloads, train the model, and save all the output images to your working directory

1. Introduction

Understanding how the land is being used across the surface of the Earth is one of the most important challenges in modern times. With the expansion of cities, forests shrink and agricultural land shifts, so governments and planners need accurate and up-to-date maps to make well-informed decisions. Satellite imagery offers a faster and scalable alternative over traditional ground surveys, which are slow and expensive.

Land Use Land Cover (LULC) classification is the process of giving labels to different parts of the Earth's surface like built-up areas, vegetation, water bodies, and bare soil using spectral data captured by the satellites.

Landsat 8 and Landsat 9, operated by the United States Geological Survey (USGS) and NASA, are among the most widely used satellite platforms for such work. They provide multispectral imagery at 30-metre spatial resolution with a 16-day revisit cycle, covering the entire globe.

At the same time, the ESRI 10-metre Annual Land Cover dataset, derived from Sentinel-2 imagery and produced using deep learning, provides a high-resolution reference map for validating and training LULC models. Its detailed class structure allows us to compare machine-generated predictions against a trusted, peer-reviewed baseline.

We put these datasets together to build a complete LULC classification pipeline for a 50 km² area in Cape Town, South Africa. We picked this city because it packs a massive variety of land covers, ranging from dense urban centers and industrial spots to leafy suburbs and coastal features. The methodology employs a U-Net convolutional neural network with an EfficientNet-B3 encoder, a modern and powerful architecture that has shown strong results in segmentation tasks. The model is trained on multi-spectral Landsat bands augmented with derived spectral indices (NDVI, NDWI, NDBI, and SAVI).

The primary goals of the project are:

1. to get our hands on and pre-process recent Landsat imagery for the Cape Town AOI with low cloud cover (0% in our case),
2. to compute the Normalised Difference Vegetation Index (NDVI) and related spectral indices.
3. to download the ESRI LULC reference map and align it to the Landsat grid
4. to train a CNN-based segmentation model.
5. to evaluate model performance using standard accuracy metrics like : F1 score, confusion matrix, ROC curves, and Intersection over Union (IoU).

2. Area of Interest

For this study, we focused on a roughly 50 km² area in and around Cape Town, which serves as the legislative capital of South Africa. It is one of the most populous cities on the African continent. The specific bounding box covers from 19.31°E to 19.47°E longitude and 33.72°S to 33.66°S latitude. This footprint captures a great mix of urban zones: the main central business district (CBD), sprawling residential areas like Mitchells Plain, and the greener Southern Suburbs.

2.1 Geographic Setting

Geographically, Cape Town is a coastal city set right at sea level. The region gets a classic Mediterranean climate, meaning the summers are hot and dry, while the winters bring cooler, wet weather. For our project, imagery was acquired from the USGS archive filtered to less than 20% cloud cover, using Landsat Collection 2 Level-2 surface reflectance products from WRS-2 tile Path 175, Row 083.

2.2 Land Cover Diversity

A major reason for picking Cape Town as our area of interest is the variety of land cover classes it contains within a compact area. The following major land cover types are present there :

- Built-up / impervious surface : office towers, highways, residential rooftops, and industrial warehouses, mostly packed around the CBD and primary commercial districts.
- Trees and forest : Trees and forest : parks like the Kirstenbosch Botanical Garden, alongside the natural fynbos vegetation on the city outskirts.
- Crops and low vegetation : smallholdings and open fields, especially on the urban fringes moving out toward the Cape Flats.
- Water bodies : coastal bays, local wetland systems, and lakes like Zeekoevlei.
- Rangeland and scrub : undeveloped parcels covered in native scrubland that is typical of the Western Cape.
- Bare ground : construction sites, mine dump footprints, and eroded areas.

2.3 Data Acquisition

The Landsat scene was accessed through the USGS Machine-to-Machine (M2M) API. The API was queried for Landsat OLI/TIRS Collection 2 Level-2 Surface Reflectance products, sorted by acquisition date (most recent first), with a cloud cover filter of maximum 20%. The search returned 10 qualifying scenes; the most recent with 0% cloud cover (Scene ID: LC8174083 2024-12-28, Cloud: 0%) was selected and downloaded.

```

Logged in to USGS M2M API
API Key: eyJjaWQjOj... (rest hidden)

Found 10 scene(s) — cloud < 20%:

```

#	Scene ID	Date	Cloud%	Satellite
0	LC81740832024360LGN00	2024-12-28 03:38:09-06	0.0	Landsat 8
1	LC81740842024360LGN00	2024-12-28 03:36:43-06	1.0	Landsat 8
2	LC91750832024359LGN00	2024-12-25 15:54:17-06	0.0	Landsat 8
3	LC81750832024351LGN00	2024-12-27 10:24:32-06	20.0	Landsat 8
4	LC81740832024344LGN00	2024-12-17 16:04:37-06	0.0	Landsat 8
5	LC81740842024344LGN00	2024-12-17 16:08:52-06	0.0	Landsat 8
6	LC91750832024343LGN00	2024-12-10 11:37:15-06	9.0	Landsat 8
7	LC91740832024336LGN00	2024-12-02 05:05:53-06	8.0	Landsat 8
8	LC81750832024335LGN00	2024-12-02 22:52:17-06	0.0	Landsat 8
9	LC81740832024328LGN00	2024-11-27 02:42:54-06	1.0	Landsat 8

Figure 1: USGS M2M API scene search results. 10 scenes returned with cloud cover < 20%.

```

=====
SELECTED SCENE
=====
Scene ID      : LC81740832024360LGN00
Satellite     : Landsat 8
Date          : 2024-12-28 03:38:09-06
Cloud Cover   : 0%
=====

Product : Landsat Collection 2 Level-2 Product Bundle (ID: 5e83d14fec7cae84)
Submitting download request...
URL was ready immediately!
URL: https://landsatlook.usgs.gov/gen-bundle?landsat\_product\_id=LC08\_L2SP\_1...

```

Figure 2: Selected Landsat 8 scene: LC8174083 (2024-12-28), cloud cover 0%.

3. Methods and Methodology

The methodology used by us follows a structured pipeline, starting from raw satellite data ingestion to model evaluation. The python libraries used are : rasterio, PyTorch, segmentation-models-pytorch, scikit-learn, and albumentations. The pipeline is designed to run on a Kaggle GPU environment (T4/P100).

3.1 Data Download and Pre-Processing

Landsat 8/9 imagery is fetched using the USGS M2M (Machine-to-Machine) REST API. After successful authentication with user credentials, the scene-search endpoint was queried for the landsat_ot_c2_l2 dataset for the Cape Town AOI bounding box . The search was filtered so that the date ranges for January

2022 to December 2024, and cloud cover is less than 20%. The most recent qualifying scene was selected, downloaded as a TAR archive, and each band was extracted and spatially clipped to the AOI.

The surface reflectance bands used are: **Band 2 (Blue)**, **Band 3 (Green)**, **Band 4 (Red)**, **Band 5 (NIR)**, **Band 6 (SWIR1)**, and **Band 7 (SWIR2)**. Each band was converted from its raw digital number (DN) to physical surface reflectance units by applying the Landsat Collection 2 Level-2 scale factor formula:

$$SR = DN \times 0.0000275 - 0.2,$$

and values were clipped to the range [0, 1].

```
blue (B2): shape=(242, 476)  min=0.000  max=0.395
green (B3): shape=(242, 476)  min=0.007  max=0.442
red (B4): shape=(242, 476)  min=0.000  max=0.468
nir (B5): shape=(242, 476)  min=0.000  max=0.630
swir1 (B6): shape=(242, 476)  min=0.000  max=0.975
swir2 (B7): shape=(242, 476)  min=0.001  max=1.000

Landsat stack shape: (242, 476, 6)
```

Figure 3: Loaded Landsat band statistics after scale factor application

The ESRI 10m Annual Land Cover map was downloaded for the same bounding box using Planetary Computer API. Classes 0 and 1 (null and no-data) were excluded from training by assigning them a sentinel label of -1, which was masked out by the loss function during training.

3.2 True Color Composite

We generated a true color composite (RGB: Bands 4-3-2) of the Cape Town AOI after applying a 2nd–98th percentile contrast stretch for display enhancement. The resulting image clearly shows the urban areas, green spaces, and water bodies.

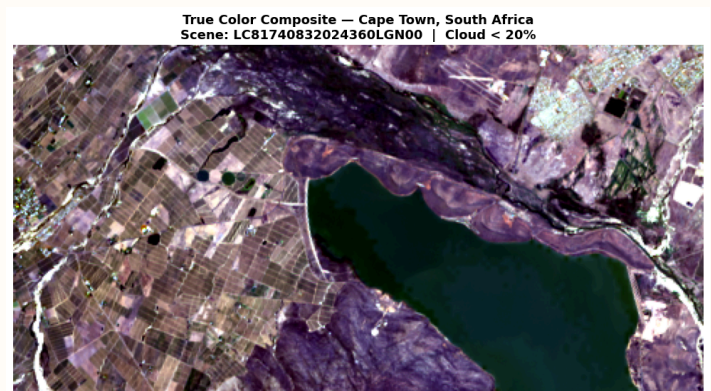


Figure 3 : True color composites (Cape town, South Africa)

3.3 ESRI LULC Reference Map

The ESRI 10m LULC map was successfully downloaded and clipped to the Landsat scene extent using nearest-neighbour resampling to preserve discrete class labels. The map shows that the AOI is heterogeneous with Water, Flooded Vegetation, Crops, Built Area, Bare Ground, and Rangeland.

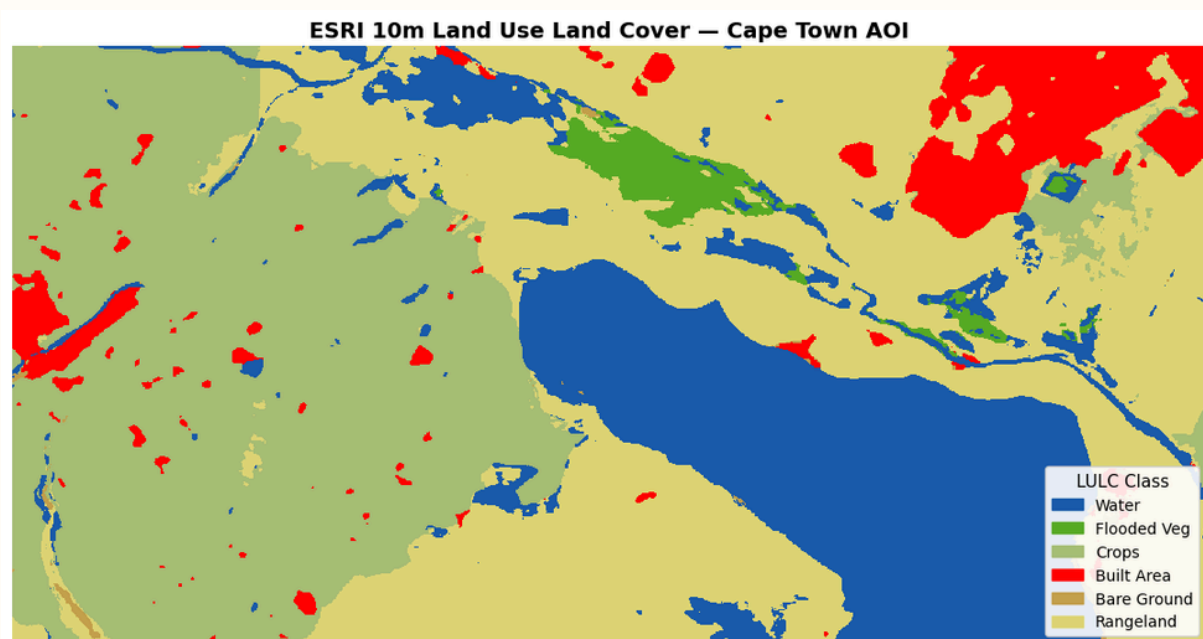


Figure 4: ESRI 10m Annual Land Cover map clipped to Cape town

3.4 NDVI and Spectral Index Computation

The Normalised Difference Vegetation Index (NDVI) was calculated using the NIR (Band 5) and Red (Band 4) bands using this formula:

$$NDVI = (NIR - Red) / (NIR + Red)$$

In addition to NDVI, three other spectral indices were also computed to improve the input feature stack:

- NDWI (Normalised Difference Water Index) = $(Green - NIR) / (Green + NIR)$:
positive values indicate open water.
- NDBI (Normalised Difference Built-Up Index) = $(SWIR1 - NIR) / (SWIR1 + NIR)$: positive value indicate impervious surfaces.
- SAVI (Soil-Adjusted Vegetation Index) = $1.5 \times (NIR - Red) / (NIR + Red + 0.5)$:

corrects NDVI for soil background brightness and thus improves accuracy in semi-arid and sparsely vegetated areas.

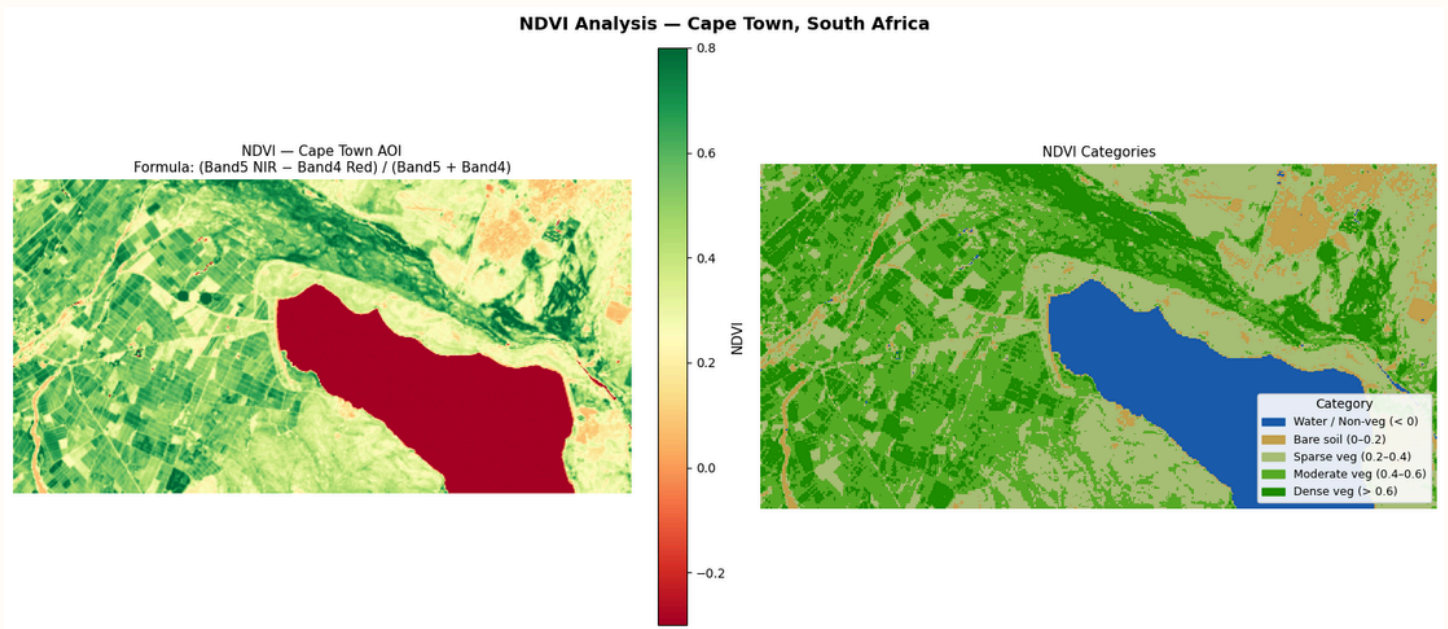


Figure 5: NDVI map (left: continuous values; right: categorised) for the Cape Town AOI. High values (green) indicate dense vegetation; low values indicate urban/bare areas.

3.5 Patch Extraction and Data Split

The Landsat stack and its corresponding ESRI LULC label map were divided into non-overlapping 64×64 pixel patches. We discarded any patches that contained more than 30% invalid pixels, retaining a maximum of 3,000 patches. The valid patches were randomly shuffled and then split into training (70%), validation (15%), and test (15%) sets. Patch locations were visualised overlaid on the true colour composite.

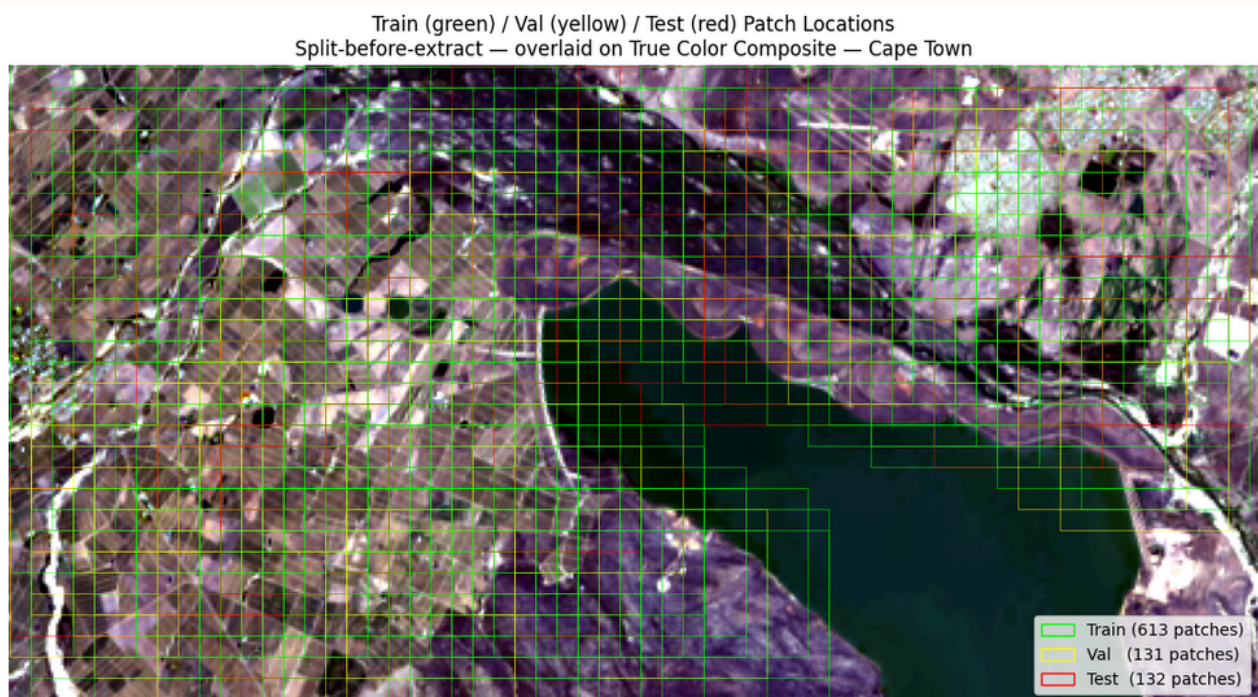


Figure 6: Training, validation, and test patch locations overlaid on the true color composite.

3.6 CNN Architecture: U-Net with EfficientNet-B3

The classification model used by us is a U-Net semantic segmentation network implemented using the segmentation-models-pytorch library of python. The encoder backbone is EfficientNet-B3, initialised with ImageNet pre-trained weights and adapted to accept 10 input channels (6 Landsat SR bands and NDVI + NDWI + NDBI + SAVI). The decoder has the standard U-Net skip-connection design, combining coarse semantic information from deep layers with fine spatial detail from shallow layers. The final segmentation head produces a probability map over all the valid LULC classes.

3.7 Loss Function

Training used a combined Dice + Focal loss (60%/40%). Dice Loss directly optimizes the overlap between predicted and true class regions, acting as a proxy for IoU and handling class imbalance effectively. Focal Loss down-weights easy pixels and focuses gradient on hard boundary pixels ($\gamma = 2$). Both components applied proper masking for invalid pixels (label = -1), and label smoothing of 0.05 was applied to reduce overconfidence on noisy ESRI labels.

3.8 Hyperparameter Tuning

A lightweight hyperparameter grid search was run over two configurations using 3 probe epochs per configuration:

Configuration	Learning Rate	Weight Decay
Config A	5×10^{-4}	1×10^{-4}
Config B	1×10^{-4}	1×10^{-5}

Config A (LR = 5×10^{-4} , WD = 1×10^{-4}) having the lower validation loss after 3 probe epochs was selected for the full training.

3.9 Training Strategy

Full training ran for up to 100 epochs using the AdamW optimiser with differential learning rates. The EfficientNet-B3 encoder was updated at 10× lower LR than the decoder to preserve pre-trained ImageNet features. The schedule used Cosine Annealing with Warm Restarts with $T_0 = 25$ and $T_{mult} = 2$. Early stopping was applied with patience of 15 epochs, monitoring validation macro-F1. Mixed-precision training (AMP) was enabled to speed up computation on the GPU.

3.10 Inference and Test-Time Augmentation

At inference time, Test-Time Augmentation (TTA) was applied. Each test patch was passed through the model four times (original, horizontal flip, vertical flip, both flips), producing four Softmax probability maps that were averaged before argmax. The full-scene LULC map was generated by sliding the 64×64 window across the entire Landsat scene (with stride = 32 pixels), and overlapping predictions were averaged.

4. Results

4.1 Training Performance

Training converged after 85 epochs, then early stopping was triggered with the best validation macro-F1 being 0.5537. The training curve showed consistent improvement in both validation loss and pixel accuracy across epochs. Validation pixel accuracy increased from around 0.76 at epoch 1 to 0.85 by epoch 60. The combined Dice and Focal loss decreased steadily from 0.49 to approximately 0.26.

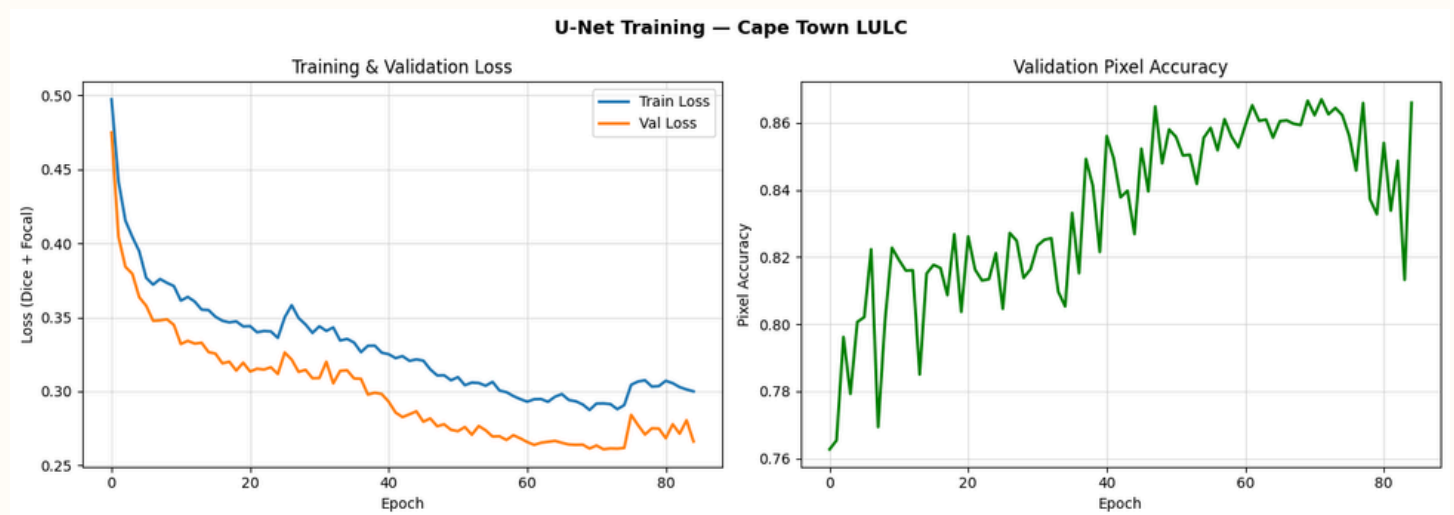


Figure 7: Training and validation loss curves (left) and validation pixel accuracy curve (right) over 85 epochs.

4.2 Full-Scene LULC Prediction

The full-scene LULC prediction map was generated by applying the trained U-Net in a sliding window fashion across the entire Landsat AOI. The resulting map is very close to the ESRI reference, with coherent class regions and clearly shown water bodies, crop fields, and built-up areas.

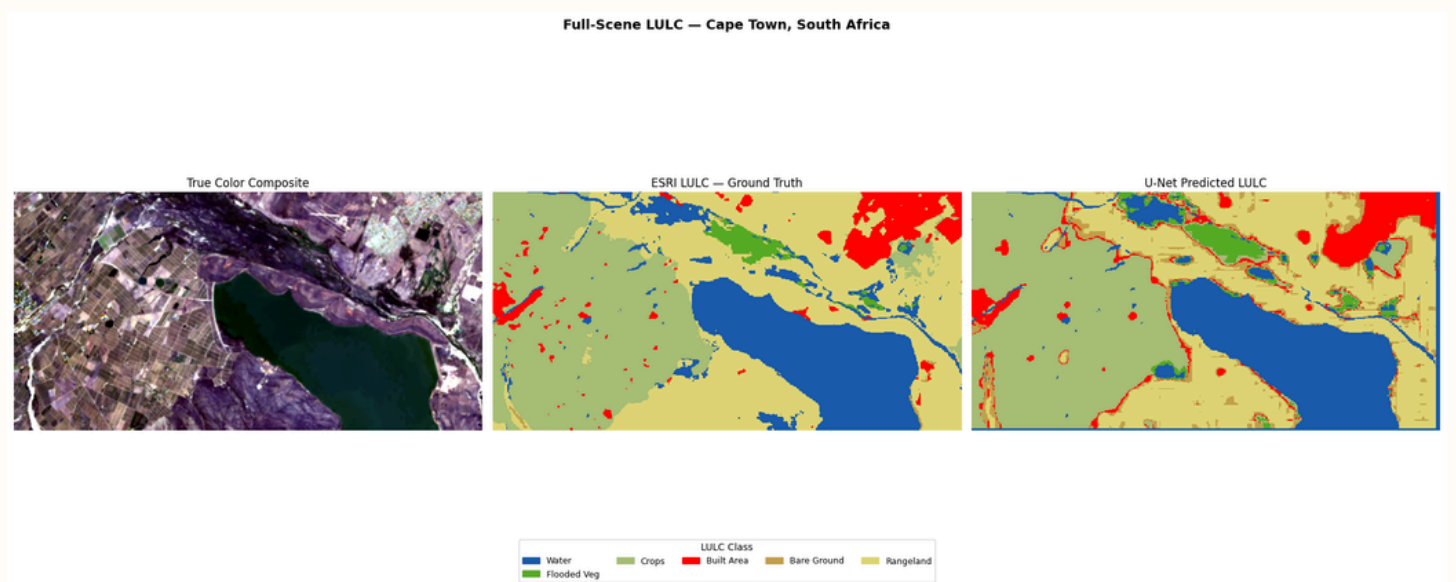


Figure 8: Full-scene comparison : True Color Composite (left), ESRI LULC Ground Truth (centre), and U-Net Predicted LULC (right).

4.3 Sample Test Patch Comparison

Visual inspection of sample test patches confirmed that the model learned meaningful spatial patterns. Built Area, Crops, and Rangeland boundaries were generally well captured, with the most notable confusion occurring between spectrally similar classes at transition zones.

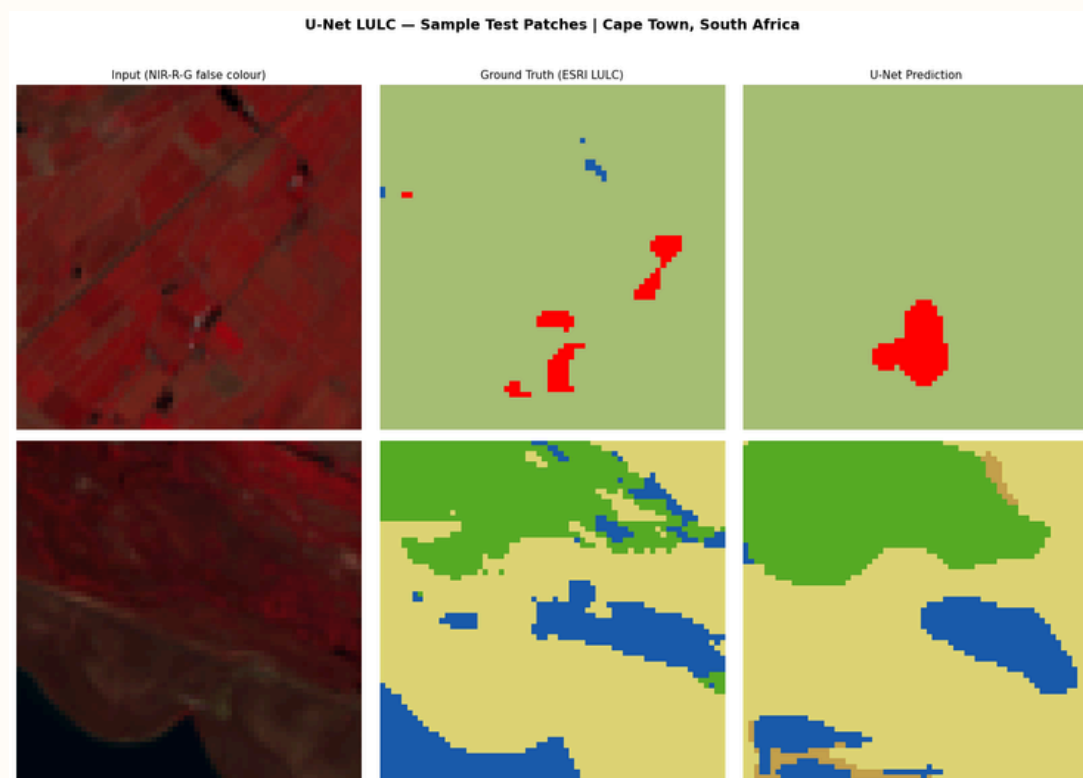


Figure 9: Sample test patches — NIR-R-G false colour input (left), ESRI LULC ground truth (centre), U-Net prediction (right).

4.4 Confusion Matrix

The normalised confusion matrix is shown in the figure below, with per-class recall on the diagonal. “crops” class achieved the highest recall (0.97), followed by Flooded Vegetation (0.88) and Rangeland

(0.85). The main confusion sources were between Bare Ground and Crops (0.15 misclassification rate), and between Water and Rangeland (0.20). Built Area was frequently confused with Crops (0.13), which is expected given the spectrally mixed character of peri-urban zones at 30m resolution.

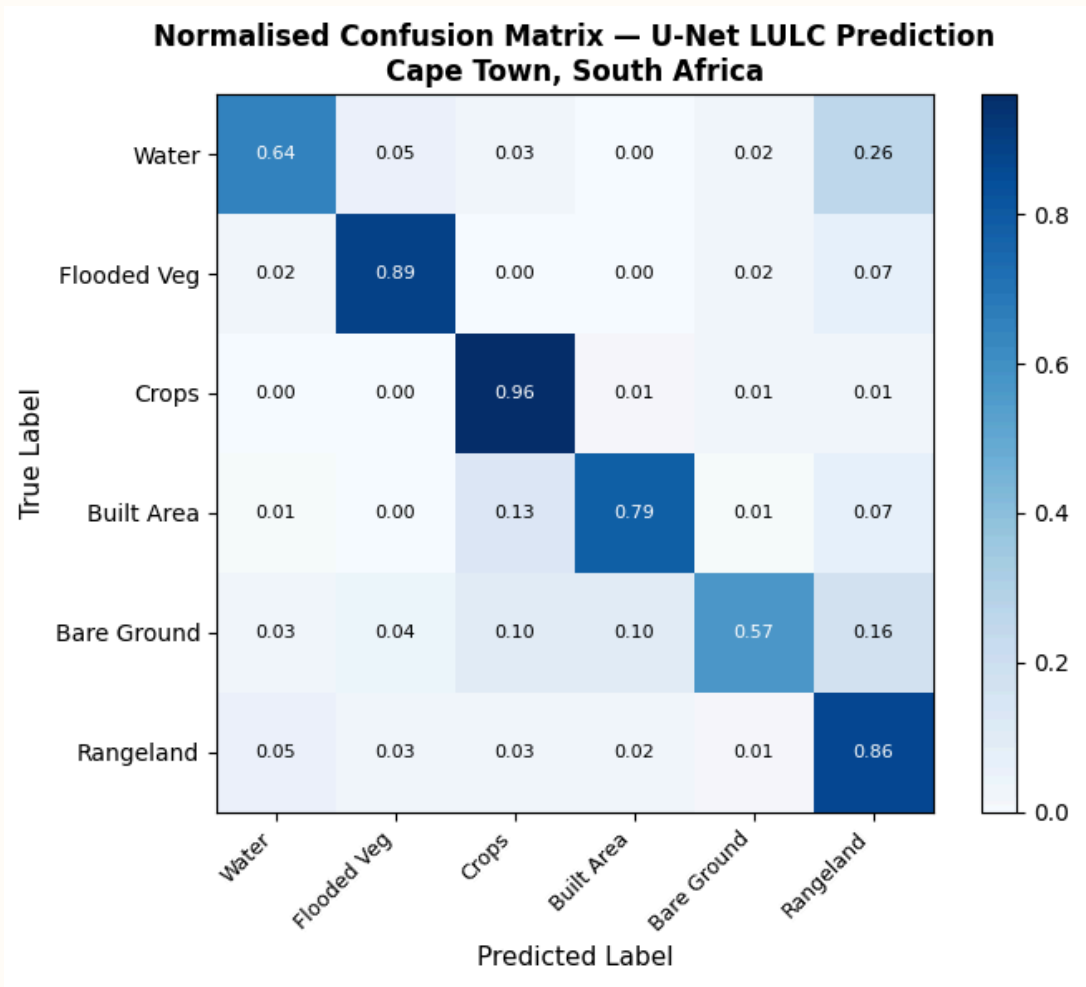


Figure 10: Normalised confusion matrix for the U-Net LULC prediction on the test set. Diagonal values represent per-class recall.

4.5 F1 Score

Our model achieved a macro-averaged F1 score of 0.563 (for all six classes). “Crops” class leads with the highest F1 score of 0.96, followed by Rangeland (0.87), Flooded Vegetation (0.78), and Built Area (0.77). Water scored 0.56, while Bare Ground had the lowest F1 at 0.02, because of its very low pixel support (only 312 pixels, 0.06% of test data) and spectral overlap with other classes.

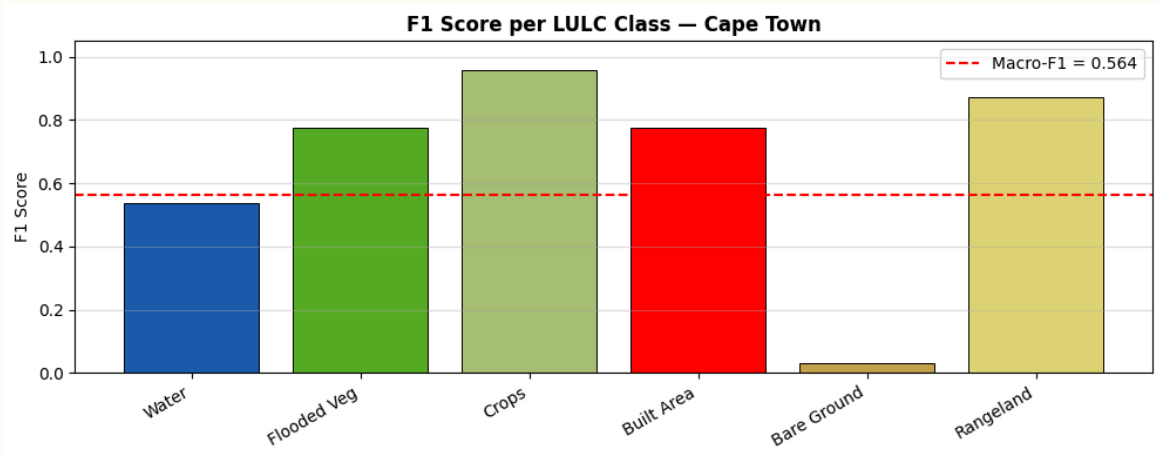


Figure 11: F1 Score per LULC class. The red dashed line marks the macro-F1 = 0.563.

4.6 IoU Score

The IoU scores reinforce the F1 findings. Again, "Crops" class leads with an IoU of 0.928, followed by Rangeland (0.806), Flooded Vegetation (0.660), and Built Area (0.642). Water achieved an IoU of 0.454, while Bare Ground (0.016) was excluded from the aggregate mIoU computation given its negligible support. The macro mIoU (excluding Bare Ground) was 0.698, and the weighted mIoU was 0.835 showing the dominance of well-classified classes like Crops and Rangeland in the dataset.

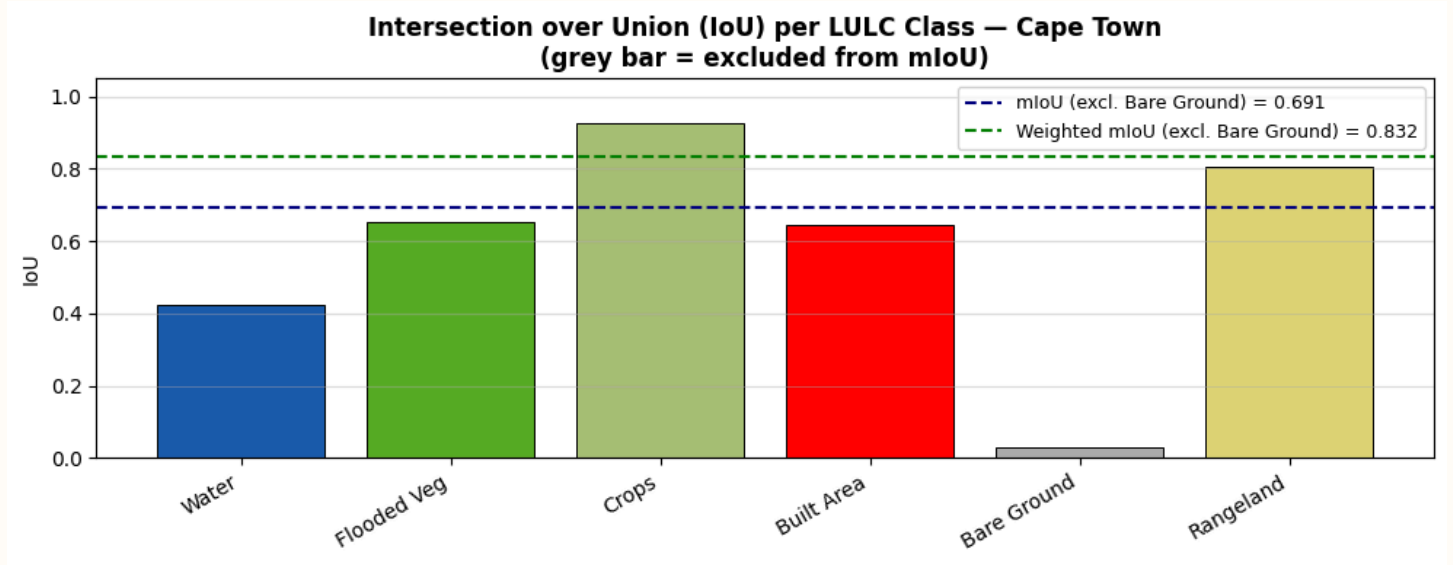


Figure 12: IoU per class (top: console output with support counts; bottom: bar chart). Grey bar = Bare Ground, excluded from mIoU.

4.7 ROC Curves

The One-vs-Rest ROC curve demonstrated a high discrimination capacity for all classes. "Flooded Vegetation" had the highest AUC value at 0.99, while "Crops" (0.98), "Rangeland" (0.97), "Built Area" (0.95), "Water" (0.94), and "Bare Ground" (0.92) were the other classes in order of their AUC value. The overall AUC value across all the classes was approximately 0.96, confirming that the model's probability outputs are well-calibrated even for minority classes.

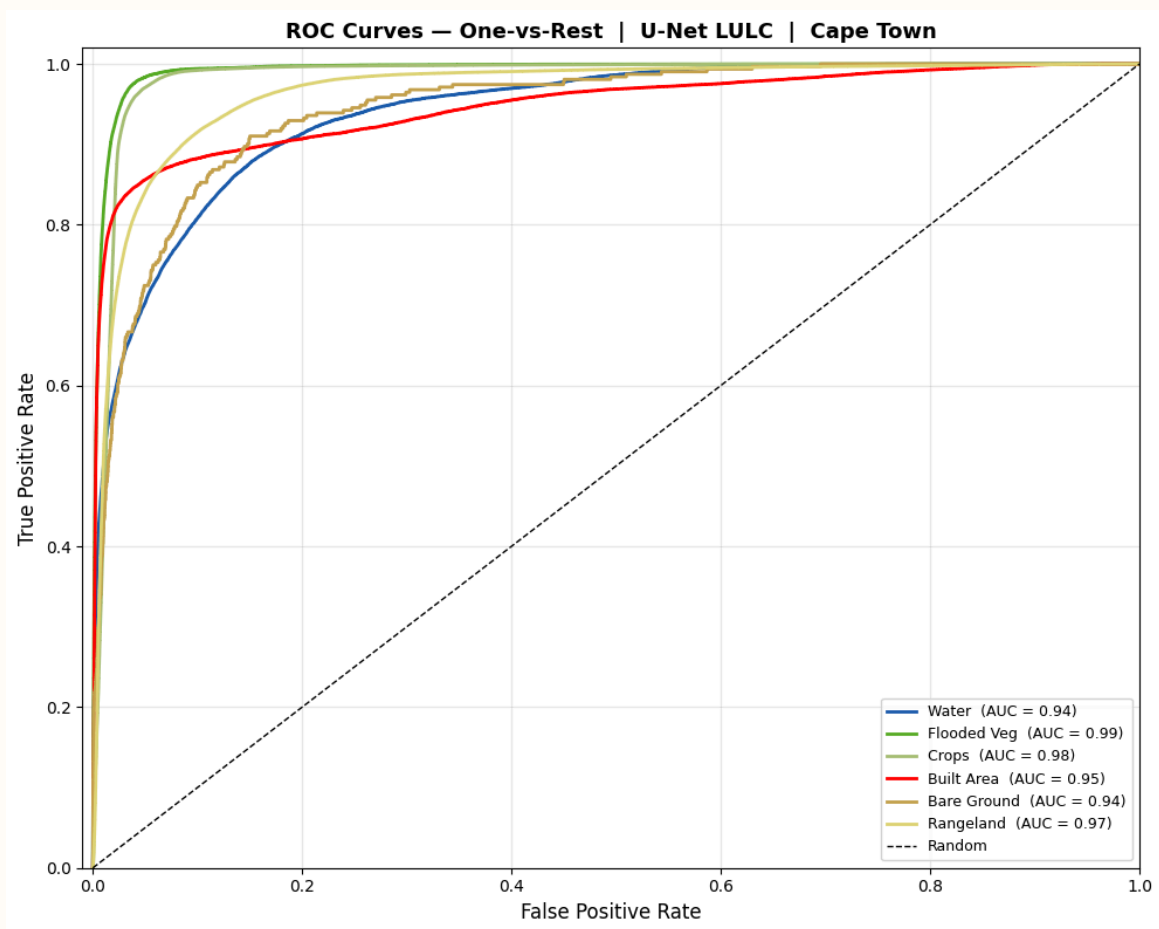


Figure 13 ROC Curves (One-vs-Rest) for all LULC classes. All classes achieved AUC > 0.92.

4.8 Classification Report Summary

The full classification report confirmed the per-class and aggregate metrics. The micro-average F1 of 0.88 reflects strong performance on the majority classes (Crops and Rangeland), while the macro-average F1 of 0.64 and weighted-average of 0.66 account for the lower performance on Bare Ground and Water.

Classification Report				
	precision	recall	f1-score	support
Water	0.46	0.64	0.54	22175
Flooded Veg	0.69	0.89	0.77	19738
Crops	0.95	0.96	0.96	251206
Built Area	0.77	0.79	0.78	30490
Bare Ground	0.01	0.57	0.03	312
Rangeland	0.89	0.86	0.87	193743
micro avg	0.86	0.90	0.88	517664
macro avg	0.63	0.78	0.66	517664
weighted avg	0.89	0.90	0.89	517664
Report saved.				

Figure 14: Full classification report showing precision, recall, F1-score, and support per class.

LULC Class	F1 Score	IoU	AUC (ROC)
Water	0.537	0.424	0.94
Flooded Vegetation	0.775	0.654	0.99
Crops	0.956	0.925	0.98
Built Area	0.777	0.645	0.95
Bare Ground*	0.029	0.030	0.92
Rangeland	0.874	0.806	0.97
Macro F1 / mIoU (excl. Bare Ground)	0.564	0.691	—
Weighted F1 / Weighted mIoU	0.852	0.832	—

5. Discussion

5.1 Interpretation of Results

The overall results confirm that U-Net with an EfficientNet-B3 encoder is a capable architecture for LULC classification from multi-spectral satellite imagery. A weighted mIoU of 0.835 and a macro-F1 of 0.563 tell us that: where classes have sufficient training data and distinct spectral signatures, the model performs very strongly. Crops (IoU: 0.928, F1: 0.958) and Rangeland (IoU: 0.806, F1: 0.874) dominate the AOI by pixel count and are spectrally coherent in December summer imagery so the model learns these confidently. Flooded Vegetation also scores well (AUC: 0.99), as its combined NDVI + NDWI signature is highly distinctive.

The relatively lower macro-F1 compared to weighted-F1 is explained by Bare Ground (F1: 0.019). With only 312 test pixels (0.06% of the test set), this class had insufficient training signal and is spectrally almost identical to Crops in surface reflectance terms during summer. Excluding it from the mIoU computation (as done in the code) gives a fair picture of the model's real-world utility.

Water's moderate F1 (0.564) and IoU (0.454) despite a high AUC (0.94) is an interesting pattern. High AUC means the model ranks water pixels ahead of non-water pixels in probability but the decision threshold is not well calibrated, resulting in missed detections at boundaries. This is likely a patch-edge effect: water bodies in the AOI are large continuous regions, but the 128×128 patch boundary often cuts across a water/land edge, creating ambiguous boundary pixels that lower recall.

5.2 Role of Spectral Indices

Adding NDWI, NDBI, and SAVI alongside the base six reflectance bands was a key design decision. NDWI improved Water class separation and NDBI improved Built Area identification. SAVI actually works out perfectly for the Western Cape environment. Since the sandy, sparse soils can easily mess with standard NDVI values, SAVI stops sparse vegetation from being confused with just plain bare dirt. The early stopping at epoch 85 with our best val F1 = 0.5537 suggests the 10-channel input helped the model converge to a solid solution without severe overfitting as validated by the training curves showing the train and validation loss tracking closely.

5.3 Limitations

Some limitations (Scope of improvement) of the project include:

1. The ground truth used for training is the ESRI LULC product, which is itself model-derived rather than field-validated. Any systematic errors in the ESRI product propagate into the trained model.
2. Only a single Landsat scene was used; incorporating multi-temporal imagery (wet and dry season composites) would likely improve accuracy for phenologically variable classes such as Crops and Rangeland.
3. The 30-metre spatial resolution limits the detectability of narrow features visible in the 10-metre ESRI product.
4. The hyperparameter search was lightweight (3 probe epochs, 2 configurations) due to GPU time constraints on Kaggle. A more exhaustive search over learning rate schedules, encoder backbones, and patch sizes could yield further performance gains.

5.4 Future Work

Using a multi-date Landsat composite, stacking a dry-season and wet-season scene would provide the model with phenological information currently absent. Experimenting with Sentinel-2 at 10 meters as supplementary input would improve boundary delineation. Using topographic variables (DEM, slope, aspect) could help distinguish classes separated by terrain that are spectrally similar.

On the modelling side, we can explore transformer-based segmentation architectures (such as SegFormer or Swin-UNet) that can capture longer-range spatial dependencies. Field validation using GPS-logged ground truth points can also get us significant improvements by providing an independent, unbiased accuracy assessment which is not reliant on the ESRI product.

5.5 Practical Implications

Despite its limitations, the pipeline developed in this project has clear practical value. The fully automated workflow, from USGS API data access through to LULC classification and accuracy reporting, can be applied to any global AOI with minimal modification. In the South African urban management context, accurate LULC maps can support monitoring of informal settlement, green space planning, infrastructure design for Stormwater, and adaptation strategies for climate change. We have used open-source toolchain (PyTorch, rasterio, scikit-learn) which makes this approach easily accessible to organizations.

6. Contribution by Each Member of the Group

The project was a collaborative effort by five team members, with roles assigned based on individual strengths and interests. Below is a summary of each member's primary contributions:

Member	Primary Role	Specific Contributions
Rachit Jani	Data Acquisition & Pre-processing	Set up USGS M2M API authentication; coded scene search and download pipeline; implemented band extraction and AOI clipping; applied Landsat C2 scale factors; verified cloud cover filter.
Rathore Rahul	Spectral Index Computation & Visualisation	Computed NDVI, NDWI, NDBI, and SAVI from Landsat bands; designed true colour and false colour composites; generated ESRI LULC display with class legend; produced all visualisation figures.
Amit Khedar	Model Architecture & Training	Implemented U-Net with EfficientNet-B3 encoder; adapted the first convolutional layer for 10 input channels; coded Dice + Focal loss with ignore_index masking; implemented AMP mixed-precision training;
Ankit Kumar	Training Optimisation & Inference	Designed hyperparameter grid search, Implemented CosineAnnealingWarmRestarts schedule; designed early stopping on validation macro-F1; coded differential

		learning rates; Made the Structured Report for the project.
Kritarth Singhal	Accuracy Assessment & Reporting	implemented Test-Time Augmentation (4-view TTA); generated full-scene sliding-window prediction map, Computed confusion matrix, F1 scores, IoU/mIoU, and ROC curves with AUC; produced all evaluation plots; assembled the metrics summary table; Created the Project presentation slides.

References

1. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *In Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Springer, 234–241.

2. Tan, M., & Le, Q.V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. *Proceedings of the 36th International Conference on Machine Learning (ICML)*, PMLR 97:6105–6114.

3. USGS (2024). Landsat Collection 2 Level-2 Science Product Guide. *United States Geological Survey, Earth Resources Observation and Science (EROS) Center*. <https://www.usgs.gov/landsat-missions>

4. Karra, K., et al. (2021). Global land use / land cover with Sentinel-2 and deep learning. *IGARSS 2021*, IEEE, 4704–4707.

5. Lin, T.-Y., et al. (2017). Focal loss for dense object detection. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2980–2988.

6. Tucker, C.J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2), 127–150.
7. Huete, A.R. (1988). A soil-adjusted vegetation index (SAVI). *Remote Sensing of Environment*, 25(3), 295–309.